

Complex Project Management

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Abstract. Managing complex projects is always a challenge. During a project's life cycle, technical, cost, schedule and political complexities can arise. These challenges may include design requirement changes, resource shortages, changes in government regulations, problems with subcontractors, delays in other interdependent projects, which require detailed project plans, particular skill sets, flexibility and constant adaptability. More importantly, organizations need methods to help managers address complex situations. System Dynamics is a method that can be applied to different areas of knowledge, especially when the system is relatively complex and the behavior of key variables is difficult to predict. This paper describes the use of System Dynamics to help project managers conduct the management process and face uncertainties and complexities. Lessons learned from a Brazilian Air Force Project Management Office - PMO are presented to illustrate insights and key lessons for decision makers.

Keywords. System Dynamics. Knowledge Management. System Thinking. Dynamic Models. Complex Projects. Lessons Learned.

Introduction

In the late 1980s, Empresa Brasileira de Aeronautica SA (Embraer) and Aermacchi launched [the AMX project](#), a joint Italian/Brazilian program to create a lead-in fighter trainer and light attack aircraft. The result was a capable jet, especially in the light attack role, known as the AMX fighter bomber which entered into service in 1990 and had the Italian and Brazilian Air Forces as the main contractors [1].

At that time, the Brazilian Air Force Command (COMAER) created the Coordinating Committee of Combat Aircraft Program (COPAC) to manage the acquisition process and coordinate the implementation of these aircrafts in Brazil. Furthermore, COPAC became the main agency responsible for coordinating activities related to modernization and acquisition of strategic aeronautical systems for Brazilian Air Force (BAF) and currently functions as a Project Management Office (PMO) in COMAER.

Presently, COPAC manages 20-23 projects with an annual budget of around US\$4 billion. Its responsibilities and activities include, among others:

- Overseeing time, cost and performance in order to adhere to contractual requirements;
- Ensuring that all work performed is both authorized and funded by contractual documentation;
- Integrating work across the operational lines, such as engineering, R&D, logistics and human resources; and

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- Ensuring that all work required is documented and distributed to all key personnel.

As these activities are carried out in these medium- and long-term projects, that is, projects lasting between 5-15 years, the management process becomes complex. There are likely to be many changes and disruptions, development of large complex systems and software, multiple integrated interfaces and interfacing with multiple complementary projects, systems and users. One tool that can help managers address these complexities is the use of System Dynamics concepts. System Dynamics is a method used to enhance learning in complex systems and can be applied to different areas of knowledge, especially when the system is relatively complex and the behavior of key variables is difficult to predict [2].

This work aims to present the use of System Dynamics concepts in managing complex projects and explore several important lessons learned in COPAC in order to share best practices and help other organizations manage their complex projects.

1. System Thinking and System Dynamics

Too often programs and projects suffer from vague and constantly changing user requirements, combined with inadequate or incomplete acceptance criteria. In this regard, COPAC has faced different project management challenges in and recognizes the necessity of new management strategies to handle these complexities. It needs tools and processes that help project managers to understand the issues, design better operating policies and guide changes in the system to improve procedures, in both simple and complex projects. The term “complex” projects refers to projects that may face challenges involving cost, schedule or performance issues, though these problems may only require constant schedule changes, requirement review and cost containment measures[3]. COPAC’s projects are complex as well as large-scale, long-term and full of uncertainties.

Besides technical and schedule challenges, funding is a constant source of concern. From not having enough to the ways in which it is phased into the project’s schedule, it requires constant contract renegotiations to introduce changes in schedule and technical requirements. These changes can reduce a project’s critical requirements and result in a suboptimal product or service. Due to the nature of its projects, COPAC also deals with political complexities. Public funding is controlled by politicians with their own priorities, ideologies and different views of projects. This requires communication strategies with politicians that highlight public interests such as national defense, scientific achievement, public utility and employment opportunities[3]. A dynamic complexity arises from the interactions between the aforementioned factors. These effects can combine, making it difficult to see how root causes and outcomes relate. Projects often get off track and grow in cost and time without stakeholders understanding the underlying reasons.

One identified tool to deal with these complexities is the practice of System Thinking. This concept helps managers to see the “big picture” of the project life cycle, and maintain focus on its budget, schedule and end user’s goals. In 2007, the Fundação Nacional da Qualidade (FNQ) conducted a virtual research with 196 Brazilian executives about System Thinking and its effective use in organizations. The results show that 45% of executives recognize the importance of this method and agree that

this knowledge should be used intensively. A mere 22% use it effectively. This research reveals a lack of knowledge in terms of how to effectively use System Thinking.

These numbers do not differ significantly in the Brazilian Air Force. This is mainly because the majority of officers receive only basic System Thinking concepts during their military training and there is a lack of the practical application of this strategic approach. To minimize these knowledge gaps, COPAC recognized the importance of practicing System Thinking along the entire project's life cycle and as a skill that managers should acquire or develop. Therefore, in 2011, COPAC started a new internal project regarding knowledge management and one of the main objectives, among others, was to develop the System Thinking practice in projects management.

The practice of system thinking needs a common language to express multiple relationships in a system. In general, we are taught to see linear relations; one cause produces one or several consequences in a linear cause and effect chain. Most of our models that try to explain a system do so through a linear view, which can bias our understanding of the system. Of course, there is not only one technical approach to understanding complexity in systems and projects. Yet there must be a starting point, an educational model to inform others in the field and, from that, develop a methodology that remains true to the context in which the process is practiced and experienced.

In COPAC, this educational model was the System Dynamics methodology and it helped managers to have a dynamic perspective on their projects, problems and issues. System Dynamics can be defined as a methodology and a mathematical modeling technique to understand the behavior of complex systems over time. This approach seeks to understand a system's structures, interconnections between all components, feedback loops and how time delays will affect the whole system and its constituent parts. One aspect of System Dynamics that helps managers to analyze complexities is the extensive use of models that can represent the relationships, time delays, behaviors and causalities of a system.

2. Mental Models

Every professional or stakeholder involved in a project has a mental model, consisting of their expectations about an issue. In other words, these mental models consist of managerial frameworks, mindsets, paradigms, conventional wisdom, and assumptions. Different mental models can be useful because they allow us to process information, access past experiences and discover new opportunities. But, they can also cause great difficulties in the decision making process and can contain psychological biases². These are the main problems in most of COPAC's projects and the consequences include reworks, schedule delays and changes in costs.

² What is Psychological Bias? Psychologists Daniel Kahneman, Paul Slovic, and Amos Tversky introduced the concept of psychological bias in the early 1970s. They published their findings in their 1982 book, "[Judgment Under Uncertainty](#)." They explained that psychological bias – also known as cognitive bias – is the tendency to make decisions or take action in an illogical way. For example, you might subconsciously make selective use of data, or you might feel pressured to make a decision by powerful colleagues. Psychological bias is the opposite of common sense and clear, measured judgment. It can lead to missed opportunities and poor decision making. (<https://www.mindtools.com/pages/article/avoiding-psychological-bias.htm>)

An example that illustrates the importances of mental models occurred during a modernization program of a Brazilian maritime patrol aircraft. Four years after the contract was signed, the main service supplier made a financial compensation claim due to losses, citing a deviation in the contract execution, caused by constant requirement changes, delayed payments and technological issues. On the other hand, BAF countered that the delays and increased costs were the result of supplier's quality issues that caused reworks and faulty deliverables. Essentially, the supplier and BAF disagreed over the delays and cost overrun because their mental models were different. The use of System Dynamics models, Causal Loop and Stock and Flow diagrams, provided a shared view to both sides, helped managers and technical personnel to understand the sources of the problems and provided alternatives to solve the contest.

Causal Loop Diagrams and Stock and Flow diagrams are part of the System Dynamics' toolbox and they were used to facilitate dynamical analysis via simulation. A detailed explanation about these tools can be found in the extensive literature, but one recommended reference is John Stearman's book, called *Business Dynamics: Systems Thinking and Modeling for a Complex World*[2].

The System Thinking and System Dynamics first helped COPAC's officers to see a particular structure underlying a particular problem in the organization. Managers start thinking in terms of the system "archetypes".

3. A System Archetype

Initially within the COMAER structure, COPAC grew fast and gained a certain credibility in coordinating and managing the main acquisition and development programs in the Brazilian Air Force. Its members developed skills and processes to manage complex projects such as the AMX fighter-bomber. Due to its quality and efficient methods, it has received support and recognition from different organizations. Officers are respected by their developed management skills and junior officers are highly motivated because of the promotion opportunities. But, eventually, systems and processes became increasingly complex and difficult to manage. With more projects under COPAC's supervision, the management burden often falls on the most experienced officers, who in turn have less time to pass on or teach their past experiences and knowledge to less experienced officers. As a consequence, more errors occur, such as incomplete requirements, faulty contracts and reworks. The consequences were costly and time consuming projects.

To resolve these issues, there were quick solutions. Senior and more experienced officers were redirected to work on problematic areas, double checking past actions and procedures. By increasing their workload but not releasing them from their daily responsibilities, they became the "heroes" or super managers acting to solve COPAC's issues. The organizational culture in the Brazilian Air Force values hard workers. There is an unwritten norm that real heroes are the officers that work over forty hours a week, stay late at the office and are available at all hours. It is a culture that produces good results, but can blind workers to certain types of problems. Officers act directly in response to the symptoms of a problem (managing errors), releasing the pressure in the organization. However, doing this, they bypass the processes that made COPAC's growth, such as the research and development (R&D) of new skills and processes to face new challenges and uncertainties in projects.

Systems Thinking provides insights that help us understand certain recurring structure patterns, called System Archetypes, that explain common management situations. Researchers have identified a dozen system archetypes [4], but one of them can be used to enlighten COPAC's historical context, known as "shift the burden".

Figure 1, shows this archetype, there are two processes and both try to adjust or correct the same problem (e.g. errors in managing projects). The symptomatic solution (top circle) solves the problem quickly, but only temporarily. The fundamental solution (bottom circle) works far more effectively, but has a delay, so it takes longer to become evident.

In shifting the burden, a side effect appears from the symptomatic solution, people rely more and more on the symptomatic solution [4], making matters worse in the long term and, at times, top managers are uninvolved due to other career demands or even retirement. The gap between a fundamental solution (e.g. train junior officers and support R&D), and the symptomatic solution (e.g. pressure senior officers to solve urgent problems), becomes worse because sufficiently experienced managers and knowledge are not retained to help those remaining in the organization.

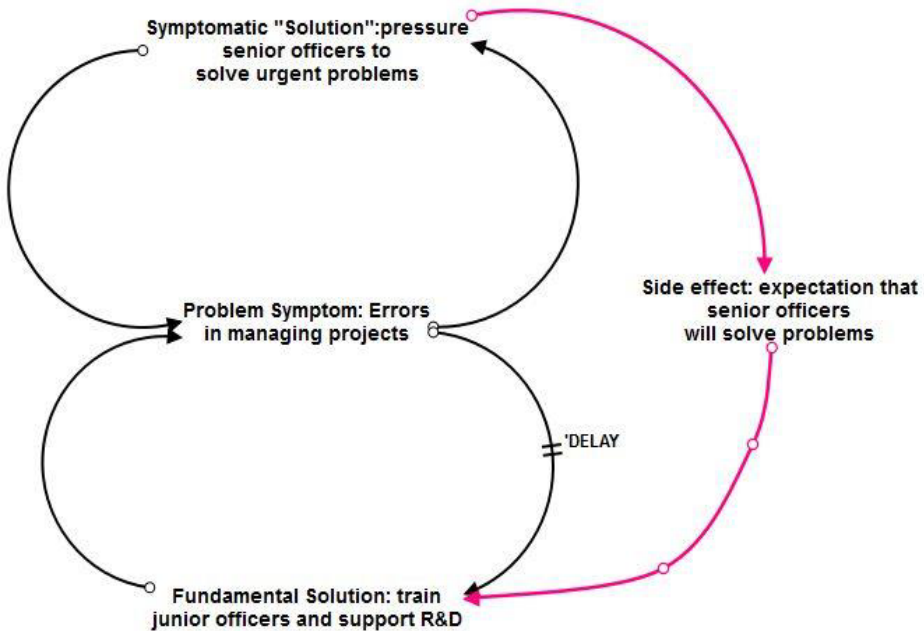


Figure 1. Shift the burden archetype, adapted from Senge [4].

In his book "The fifth discipline: The art and practice of the learning organizations" Peter Senge [4] discusses why systemic insight are not incorporated in operating policies. Even if a pilot experiment proves successful and a new approach leads to better results, widespread adoption of the approach rarely occurs. He explains that new insights fail to get put into practice because they conflict with deeply held internal images of how the world works (mental models); these images limit us to

familiar ways of thinking and testing rather than allowing us to challenge and question the world around us.

In the COPAC example, these problems came from inadequate or incomplete mental models. These models, developed over time through experience, can act as filters that screen incoming information and make us hear what already supports our existing beliefs and ways of operating. Meanwhile, any new information that does not support what we believe is discarded as wrong or not applicable. In BAF, high-rank officers believed the solution should come from increasing the workload of the senior managers.

The solution for this archetype was to improve junior officers training and the knowledge management in the organization. Basically, explaining to high-rank officers the logic of this model, what was occurring and the long-term consequences in order to support the fundamental solution (Figure 1).

In this context, the analysis of works presented by Tavares [5] and Bonotto [6], describes the knowledge production and project management's vulnerabilities in COPAC. Through these works, an important issue could be also identified: the loss of organizational knowledge due to the departure of skilled staff. So, an internal initiative was established.

4. Knowledge Management

Like other types³ of knowledge, the management of these assets is not an easy task. Organizational knowledge, whether explicit or implicit, is an abstract entity inherent to humans and is not transferred or shared with ease or spontaneity [7]. Unlike the administration of physical resources (material, human and financial), which can be measured and controlled, organizational knowledge is subjective, is not well defined and conceptualized, and lacks explicit indicators.

Different authors define Knowledge Management in different contexts and for different purposes; they conceptualize the term in different ways based on different epistemologies. Therefore, for this paper, the definition used incorporates an interdisciplinary perspective that considers the dynamic nature of Knowledge Management, and will be defined as:

The effective learning processes associated with exploration, exploitation and sharing of human knowledge (tacit and explicit) that uses appropriate technology and cultural environments to enhance an organization's intellectual capital and performance [8].

According to Nonaka and Takeuchi [9] there are two types of knowledge, tacit (implicit) and explicit knowledge. Tacit knowledge consists of one's experiences, mental models, beliefs and opinions. Explicit knowledge is a kind of knowledge that can be defined and shared easily through information technology. With these two concepts, Nonaka and Takeuchi have developed a theoretical dynamic model as shown in Figure 2, known as SECI (Socialization, Externalization, Combination and Internalization). This framework is based in studies on the Theory of Organizational Knowledge Creation [9].

³ Popular, philosophical, religious or scientific.

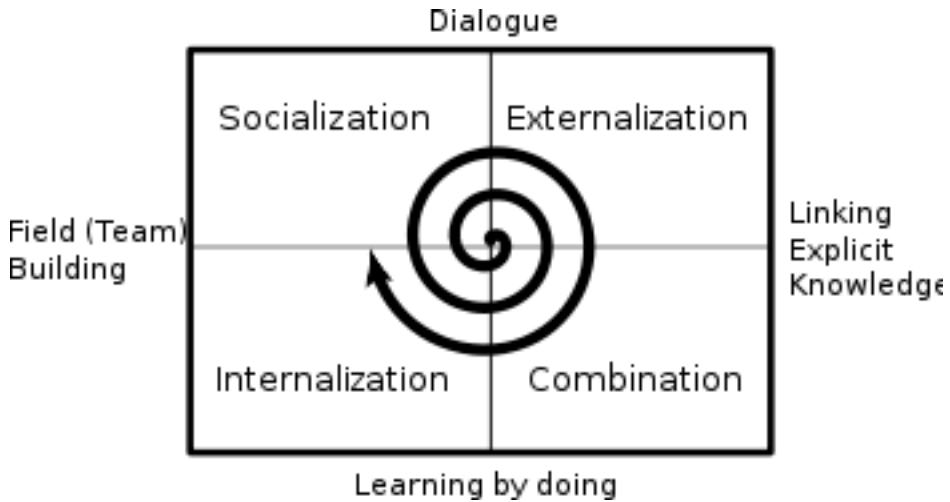


Figure 2. Nonaka and Takeuchi SECI model [9].

These two authors argue that knowledge is initially created by individuals and becomes organizational knowledge through a spiral process described by the theory (see Figure 2).

This spiral is created by the four modes of knowledge conversion through which knowledge is converted from one knowledge type to another. The modes of knowledge conversion include socialization (from tacit to tacit knowledge), externalization (from tacit to explicit knowledge), combination (from explicit to explicit knowledge), and internalization (from explicit to tacit knowledge).

The SECI model is dynamic [10]; humans play an important role and organizational networks generate knowledge through social processes of sharing, exploration and creation of tacit knowledge (stories, experiences and concepts) and explicit knowledge (raw data, bank and organized data reports).

Peter Massingham [11] also explores this issue in a case study within the Australian Ministry of Defense. In this work, the author simulates the consequences of the departure of important personnel and discusses the results of this research in the context of existing literature. In his conclusion, he presents a preliminary conceptual model in order to identify and isolate the causes and nature of the risks involved.

An important tool, in terms of knowledge management and part of COPAC's initiative, is the registry of "Lessons learned". This was an effort to create, among project managers, a habit of incorporating a process of lesson learning during the life cycle of the project, not only at its end. This process attempts to make knowledge be processed by the four modes of knowledge conversion. For instance, every week a subject or a case study is presented to managers or junior officers during a general meeting (Externalization), a topic is also created in the private network to allow discussion and exchange of experiences (Combination), the important insights are captured and registered for public research (Internalization) and communities of practice are formally recognized to promote informal meeting (Socialization).

Lessons learning means that at each key milestone, people involved with the project pause to review assumptions, evaluate changes in the internal and external

environment and review risks and estimates. In this way, knowledge is spread through the entire organization.

The goal is to adapt the project plans to any changes that will impact the final objective with the shared knowledge created by the collective experience. There is a constant need to preserve experiences and accumulated memories during the management process. This implicit knowledge, combined with explicit knowledge (e.g. policies, guidelines and manuals), builds COPAC's organizational knowledge which permeates all its activities and most of this knowledge resides in the minds of its members and leaders, from past and present.

5. Conclusions

In today's world, technology changes faster than the pace of the project. Managing complex projects is always a challenge and during the entire project life cycle, there are technical, cost, schedule and political complexities. It requires management strategies that can help managers address the uncertainties and complexities in a project.

This article presented some aspects of the use of System Dynamics and the lessons learned approach along the existence of a PMO in the Brazilian Air Force Command in order to help other organizations in their project management initiatives.

As a PMO, COPAC needs to move everyone in the same direction. However, two people may look at the same problem and derive different viewpoints based on their own experiences. Suggesting a System Dynamics approach, adding concepts of delays and feedbacks loops, is not to deny people's mental model, but instead, to help them develop and improve models with a system thinking view.

System Thinking and System Dynamics help managers to routinely question their mental models. This does not necessarily means abandoning their beliefs or past experiences, but requires them to think actively about assumptions we make about a problem, a project or a program.

Another key point to manage these projects is to understand where the complexity originates from. This demands a System Thinking view to see the whole instead of the parts and helps managers to identify the relationships and consequences of decisions made. This knowledge is an important asset and has strategic value for the whole organization.

One way to preserve this knowledge is through knowledge management initiatives, especially in the form of "Lessons learned," which acknowledges that there are takeaways can learned from each and every project, from simple insights to strategic conclusions that can be usefully applied on future projects.

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